Solution Requirements Specification

Template: Functional and Non-Functional Solution Requirements	Maximum Marks: 4 Marks
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Step 1: Functional Requirements (FR)

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

S.No	Requirement Category	Requirement Description (FoodVerse AI)	Priority
1	Authentication & Security	Zero-friction public access for live testing; API key encapsulation via server-side environment variables (.env) ensuring credentials are never exposed.	High
2	Authorization Levels	Public User role (read/analyze/generate plans), Kitchen Inspector role (batch video upload for audit), Developer/Admin role (health telemetry inspection).	Medium
3	External Interfaces	FastAPI integration with Groq Multimodal API (Qwen 3.8B), Groq Whisper Turbo API for voice audio notes, and Vercel/Render cloud web endpoints.	High
4	Transactions Processing	Multipart/form-data ingestion for high-res images up to 25MB, video streaming up to 100MB, real-time JSON validation via Pydantic v2 schemas.	High
5	Reporting & Visualization	Instant display of macronutrient distribution (calories, carbs, protein, fat), OpenCV discoloration charts, hygiene compliance grades (Grade A-F), and daily meal schedules.	High
6	Business Rules & Logic	Scientific Mifflin-St Jeor metabolic calibration; hard constraints on strict allergen exclusion (peanuts, gluten, dairy); algorithmic CV fallback if API is offline.	High
7	Compliance to Regulations	Alignment with FDA Food Safety Modernization Act (FSMA) guidelines and HACCP kitchen hygiene standards.	Medium
8	Hands-Free Voice Query	Real-time culinary query handling supporting interactive chef troubleshooting without requiring physical device interaction.	High

Step 2: Non-Functional Requirements (NFR)

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Rapid multimodal inference and computer vision processing across mobile and desktop connections.	End-to-end response time < 3.0 seconds for multimodal food analysis; < 800ms for text meal planning.
2	Scalability	Stateless FastAPI microservices architecture with asynchronous non-blocking worker pools.	Capable of sustaining up to 1,000 concurrent active client requests without degradation.
3	Security & Data Privacy	Client media files stored temporarily in volatile memory buffers with immediate deletion post-analysis.	Zero persistent storage of user photos without consent; 100% deletion of temp files in finally blocks.
4	Reliability & Availability	High uptime web services hosted on redundant cloud infrastructure with automated health-check endpoints.	99.9% monthly availability; automated CV deterministic fallback if cloud LLM encounters rate limiting.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
5	Usability & Accessibility	Clean, modern, responsive dark-mode user interface adhering to WCAG 2.1 AA accessibility guidelines.	Fully mobile-responsive layout; high contrast color palette strictly avoiding illegible or purple tones.
6	Maintainability & Modularity	Separation of concerns across distinct modular router files with full typing and automated edge-case test suite.	100% pass rate across automated edge-case validation suite (9/9 tests verified).